

DAY-07(26.07.2026)

IP ADDRESSES

- Internet Protocol Address → unique that uses
- identifies and locates the device for communication

▼ **COMPONENTS OF IP:**

- Network portion → network to which it belongs
- Host portion → individual device
- Subnet Mask (IPv4) → which part is network and host

▼ TYPES OF IP ADDRESS:

▼ BASED ON ADDRESSING SCOPE:

PUBLIC IP ADDRESS

- in every device directly access the internet
- key characteristics - Uniqueness and accessibility
- assigned by ISP

PRIVATE IP ADDRESS

- within private network and cannot access the internet without Network Address Translation
- communicate within same network
- assigned by router or DHCP server

▼ BASED ON IP VERSION:

IPv4

- 4 sets (octets)
- each octet - 8bits/1byte
- value range 0 - 255
- $2^8 = 256$ combinations
- separated by dots
- ex: **192.168.1.1**

IPv6

- 8 groups of 4 hexadecimal
- 128 bits
- each group - 16 bits/2bytes
- separated by colons
- ex: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

▼ BASED ON ASSIGNMENT:

STATIC IP ADDRESSES

- permanently assigned (network services)

DYNAMIC IP ADDRESSES

- temporarily assigned by Dynamic Host Configuration Protocol (DHCP)
- cost effective , efficient for providers, perfect for consumers

▼ BASED ON FUNCTION:

UNICAST ADDRESS

- one sender and one specific receiver
- one to one
- ex: email, files

BROADCAST ADDRESS

- one device to all
- one to all
- supported in IPv4
- ex: An ARP (Address Resolution Protocol) request uses broadcasting to find a device's MAC address on the local network.

MULTICAST ADDRESS

- one device to all selected multiple device
- one to all but selected
- only the device join the group will receive the message
- ex: live video or online conference

ANYCAST ADDRESS

- one to nearest receiver among same IP address
- based on routing distance
- primarily in IPv6 and also in IPv4
- ex: Content Delivery Networks which routes user requests to nearest data center

▼ CLASSES OF IPv4 ADDRESS:

class A → 1-126 → large networks(MNC's)

class B → 128-191 → medium-sized(large organizations)

class C → 191-223 → small networks(business, home)

class D → 224-239 → multicast groups

class E → 240-254 → experimental purpose

▼ SPECIAL IP ADDRESS:

Loopback address → 127.0.0.1 → test network connectivity within the device

Broadcast Address → allows to send the data to all devices in a network → 192.168.1.255

Multicast Address → sends data to group of devices → 233.0.0.1